Revision 1: results of the fresh-data rerun

APM-26-08-243, editor comments E3 and E4, Reviewer 1 comments 1 to 3

Magnus Johansson

Table of contents

Integrity 1
E4: Table 4 extended to B = 10 000 2
R1.2: the studentization test 4
E3 and Study 1: interval convergence from fresh data 6
Interval rule under the null against Equation 1 7
Study 2 misfit conditions from fresh data 8
The detection and false positive trade-off (section 4.3) 8
Reachability sequences (section 5.1) 9
Partial nulls: familywise error over fitting items and FDR (R1.1, R1.3) 9

The design is in `rerun_plan.md` and the code in `fresh_rerun.R`. Both dichotomous studies of the submission are regenerated with fresh data in every replication, using the same 20 item locations and misfitting items. Each replication runs one bootstrap at the largest B, and every smaller B uses its first B draws. Comparisons with the submission use the cached results of Studies 1 and 2, which drew every sample from one master dataset per condition.

Integrity

```
keys <- c("rep", "k", "n")
P1 <- bind_part(T1, "pval", keys) |> mutate(cell = paste(rep, k, n))
H1 <- bind_part(T1, "hdci", keys) |> mutate(cell = paste(rep, k, n))
P2 <- bind_part(T2, "pval", keys) |> mutate(cell = paste(rep, k, n))
H2 <- bind_part(T2, "hdci", keys) |> mutate(cell = paste(rep, k, n))

short_draws <- bind_rows(P1 |> mutate(tier = 1), P2 |> mutate(tier = 2)) |>
  distinct(tier, cell, B, B_used) |>
  filter(B_used < B) |>
  summarise(n_short = n(), max_missing = max(B - B_used), .by = tier)

data.frame(
  tier = 1:2,
  cells = c(nrow(T1$grid), nrow(T2$grid)),
  failed = c(sum(vapply(T1$res, is.null, logical(1))),
             sum(vapply(T2$res, is.null, logical(1)))),
  hours = round(c(T1$elapsed_min, T2$elapsed_min) / 60, 2),
  na_pvalues = c(sum(is.na(P1$padj_fwer)), sum(is.na(P2$padj_fwer)))
) |>
  left_join(short_draws, by = "tier") |>
  knitr::kable()
```

tier	cells	failed	hours	na_pvalues	n_short	max_missing
1	2000	0	11.02	0	11	29
2	2000	0	2.18	0	6	3

All cells completed and no p-value is missing. Where a refit failed inside a bootstrap, the largest B used the draws that succeeded, and the table shows how many replication-by-B combinations were affected and by how many draws at most.

```
# One row per replication, B, item and rule
long_rules <- function(P, H) {
  P <- P |>
    mutate(p_holm = p.adjust(p_marg, "holm"),
           p_bonf = p.adjust(p_marg, "bonferroni"),
           .by = c(cell, B))
  lab <- c(p_marg = "Marginal", p_bonf = "Bonferroni", p_holm = "Holm",
           padj_fwer = "Westfall-Young", padj_bh = "Benjamini-Hochberg",
           p_marg_pool = "Marginal, joint", padj_fwer_pool = "Westfall-Young, joint")
  LP <- P |>
    select(cell, rep, k, n, B, item, all_of(names(lab))) |>
    pivot_longer(all_of(names(lab)), names_to = "rule", values_to = "p") |>
    mutate(rule = unname(lab[rule]), flag = as.integer(p < AL)) |>
    select(-p)
  LH <- H |>
    transmute(cell, rep, k, n, B, item,
              rule = sprintf("Interval %.3f", w), flag = as.integer(code > 0))
  bind_rows(LP, LH)
}
mis_of <- function(k) T1$items$mis[seq_len(k)]
stopifnot(identical(T1$items$mis, s_grid$misfit))   # same misfitting items
L1 <- long_rules(P1, H1) |> mutate(is_mis = FALSE)
L2 <- long_rules(P2, H2) |>
  mutate(is_mis = mapply(function(k, it) it %in% mis_of(k), k, item))

per_rep <- function(L, by) {
  L |>
    summarise(n_fp = sum(flag[!is_mis]), n_tp = sum(flag[is_mis]),
              n_mis = sum(is_mis), n_fit = sum(!is_mis),
              .by = all_of(c(by, "cell", "B", "rule"))) |>
    mutate(any_fp = as.integer(n_fp > 0),
           fdp = ifelse(n_fp + n_tp > 0, n_fp / (n_fp + n_tp), 0))
}
summ <- function(L, by) {
  per_rep(L, by) |>
    summarise(R = n(),
              Det = 100 * sum(n_tp) / sum(n_mis),
              FP = 100 * sum(n_fp) / sum(n_fit),
              FWER = 100 * mean(any_fp),
              se = 100 * sd(any_fp) / sqrt(n()),
              FDR = 100 * mean(fdp),
              .by = all_of(c(by, "B", "rule")))
}
```

E4: Table 4 extended to B = 10 000

```
S1 <- summ(L1, character(0))
S1 |>
  filter(rule %in% c("Marginal", "Bonferroni", "Holm", "Westfall-Young",
                     "Benjamini-Hochberg", "Westfall-Young, joint")) |>
  select(B, rule, FWER) |>
  pivot_wider(names_from = rule, values_from = FWER) |>
  rename(Marg = Marginal, Bonf = Bonferroni, WY = `Westfall-Young`,
```

```
        BH = `Benjamini-Hochberg`, `WY joint` = `Westfall-Young, joint`) |>
  arrange(B) |>
  knitr::kable(digits = 1)
```

B	Marg	Bonf	Holm	WY	BH	WY joint
100	62.2	0.0	0.0	5.6	0.0	3.5
200	60.8	0.0	0.0	4.2	0.4	3.1
400	60.4	3.7	3.7	3.8	3.7	3.4
1000	60.6	2.6	2.6	3.4	2.7	3.2
2000	61.3	3.3	3.3	3.5	3.4	3.4
5000	60.5	3.1	3.1	3.5	3.2	3.5
10000	60.5	3.2	3.2	3.4	3.4	3.4

Binomial standard errors are at most 0.51 points per cell. The submitted Table 4, from the master-dataset study with 1 000 replications per B, read 6.8, 5.2, 4.3, 4.2 and 3.8% for Westfall-Young at B = 100 to 2000.

```
fresh_n <- summ(L1, "n") |> filter(rule == "Westfall-Young") |>
  mutate(data = "fresh")
master_n <- bind_part(s_grid, "pval", keys) |>
  filter(k == 0) |>
  summarise(any = any(padj_fwer < AL), .by = c(rep, n, B)) |>
  summarise(FWER = 100 * mean(any), .by = c(n, B)) |>
  mutate(data = "master")
bind_rows(fresh_n |> select(data, n, B, FWER), master_n) |>
  pivot_wider(names_from = B, values_from = FWER, names_prefix = "B = ") |>
  arrange(data, n) |>
  knitr::kable(digits = 1)
```

data	n	B = 100	B = 200	B = 400	B = 1000	B = 2000	B = 5000	B = 10000
fresh	150	4.6	3.4	3.6	3.0	3.6	3.4	3.0
fresh	250	6.8	5.6	5.0	4.0	4.4	4.2	4.0
fresh	500	5.0	3.4	2.4	2.4	2.2	2.6	2.6
fresh	1000	6.0	4.2	4.2	4.0	3.8	4.0	4.0
master	150	4.8	3.6	2.4	2.8	3.6	NA	NA
master	250	6.4	5.2	4.4	4.0	3.2	NA	NA
master	500	6.8	4.4	3.6	2.8	2.8	NA	NA
master	1000	9.2	7.6	6.8	7.2	5.6	NA	NA

Summary for E3. The rise at n = 1000 is gone in fresh data. The submitted Westfall-Young null rate at n = 1000 was 5.6 to 9.2% across B, against 3.8 to 6.0% in the rerun, and from B = 400 the fresh rate at n = 1000 (3.8 to 4.2%) is in the range of the other sample sizes (2.2 to 5.0%). The rise was a property of the master dataset for that condition, as suspected, and not of the sample size. Pooled over sample sizes, the master-dataset study was higher at every common B, by 0.3 to 1.2 points (5.6% against 6.8% at B = 100, 3.5% against 3.8% at B = 2000).

R1.2: the studentization test

The joint variant studentizes the observed and simulated statistics by moments computed over all B + 1 values, which removes the asymmetry that the explanation in the letter points to. If the explanation is right, the joint variant should show no excess at small B, and the two variants should agree at large B.

```
pair_j <- per_rep(L1 |> filter(rule %in% c("Westfall-Young", "Westfall-Young, joint")),
                  character(0)) |>
  select(cell, B, rule, any_fp) |>
  pivot_wider(names_from = rule, values_from = any_fp) |>
  mutate(d = `Westfall-Young` - `Westfall-Young, joint`) |>
  summarise(R = n(),
            standard = 100 * mean(`Westfall-Young`),
            joint = 100 * mean(`Westfall-Young, joint`),
            diff = 100 * mean(d), se_diff = 100 * sd(d) / sqrt(n()),
            .by = B) |>
  arrange(B)
pred <- data.frame(B = sort(unique(P1$B))) |>
  mutate(c = qnorm(1 - (1 - 0.95^(1 / 20)) / 2),
         predicted = 100 * (1 - (1 - 2 * pt(-c, B - 1))^20)) |>
  select(-c)
left_join(pair_j, pred, by = "B") |> knitr::kable(digits = 2)
```

B	R	standard	joint	diff	se_diff	predicted
100	2000	5.60	3.55	2.05	0.32	6.31
200	2000	4.15	3.10	1.05	0.23	5.63
400	2000	3.80	3.35	0.45	0.15	5.31
1000	2000	3.35	3.25	0.10	0.07	5.12
2000	2000	3.50	3.40	0.10	0.07	5.06
5000	2000	3.55	3.55	0.00	0.00	5.02
10000	2000	3.40	3.40	0.00	0.00	5.01

```
attain <- function(B) 100 * floor(AL * (B + 1) - 1e-9) / (B + 1)
L1 |>
  filter(rule %in% c("Marginal", "Marginal, joint")) |>
  summarise(rate = 100 * mean(flag), .by = c(B, rule)) |>
  pivot_wider(names_from = rule, values_from = rate) |>
  mutate(`attainable size` = attain(B)) |>
  arrange(B) |>
  knitr::kable(digits = 2)
```

B	Marginal	Marginal, joint	attainable size
100	4.76	4.52	4.95
200	4.69	4.58	4.98
400	4.63	4.58	4.99
1000	4.64	4.61	5.00
2000	4.65	4.64	5.00
5000	4.61	4.60	5.00

B	Marginal	Marginal, joint	attainable size
10000	4.60	4.59	5.00

The last column is the largest rejection probability a valid Monte Carlo test can have at that B, since p-values take the values $j/(B+1)$ and $p < .05$ admits only those below .05.

```
fig_d <- bind_rows(
  pair_j |> transmute(B, value = standard, what = "Standard studentization"),
  pair_j |> transmute(B, value = joint, what = "Joint studentization"),
  pred |> transmute(B, value = predicted, what = "Predicted, t asymmetry")
) |>
  mutate(what = factor(what, c("Standard studentization", "Joint studentization",
                               "Predicted, t asymmetry")))
ggplot(fig_d, aes(B, value, linetype = what, shape = what)) +
  geom_hline(yintercept = 5, colour = "grey60") +
  geom_line() + geom_point(size = 1.6) +
  scale_x_log10(breaks = sort(unique(P1$B)),
                labels = format(sort(unique(P1$B)), big.mark = " ")) +
  scale_shape_manual(values = c(16, 1, NA)) +
  labs(x = "Bootstrap iterations (B), log scale", y = "Familywise error (%)",
       linetype = NULL, shape = NULL) +
  theme_bw() + theme(legend.position = "bottom")
```

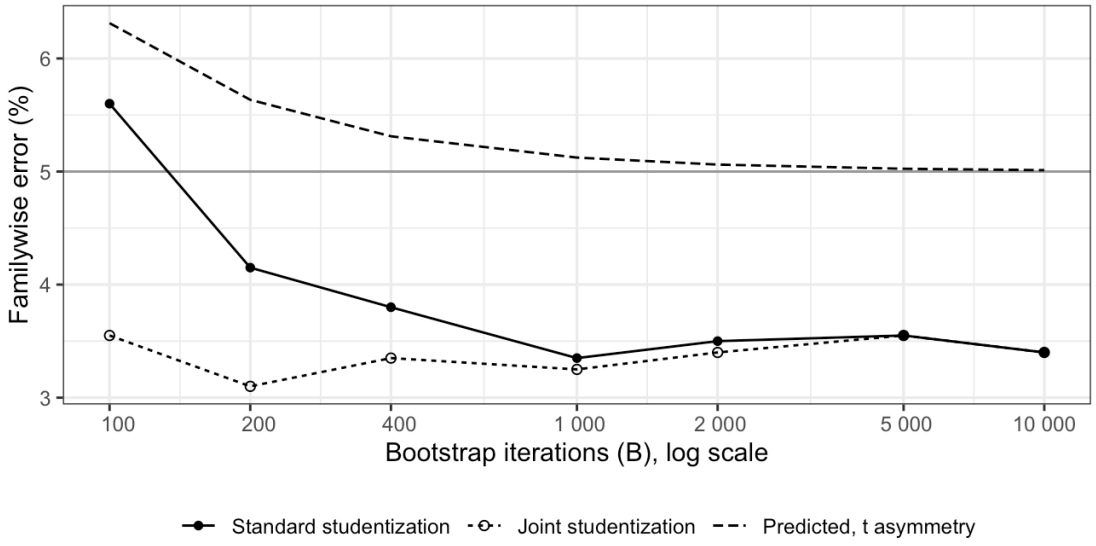


Figure 1: Westfall-Young familywise error under the null against B, 20 dichotomous items, fresh data

Summary for E4 and R1.2. The explanation is confirmed on the same draws. With standard studentization the Westfall-Young rate was 5.6% at B = 100, 4.2% at 200 and between 3.4 and 3.8% from 400 to 10 000. With joint studentization it was 3.5% at B = 100 and between 3.1 and 3.6% at every B. The paired difference was 2.05 points at B = 100 (SE 0.32), 1.05 at 200 (SE 0.23), 0.45 at 400 (SE 0.15), and 0.1 or less from 1000. The whole decline with B is therefore the studentization asymmetry, and joint studentization removes it. The t-based prediction gets the direction and the order of the decline right but understates its size at B = 100 (a predicted excess of 1.3 points over the large-B value against an observed 2.2).

What remains is a level, not a trend. Both variants settle at about 3.4%, below the nominal 5%, and the rate does not move between B = 2000 and 10 000. The per-item marginal tests show the same thing at their own level, rejecting 4.6 to 4.8% of fitting items against an attainable 5.0% at every B. A conservatism that does not depend on B cannot be Monte Carlo error. It belongs to the parametric bootstrap itself, whose reference distribution comes

from estimated parameters (the composite null of R1.2). The polytomous studies were closer to nominal (4.2 to 5.0% from B = 400 in the cached results), so the size of the conservatism depends on the item format or set.

Consequences. The letter's E4 and R1.2 responses can state the explanation as tested rather than proposed. Joint studentization makes the first step of the step-down an exact Monte Carlo test with respect to the fitted model, and in these data it made the error rate independent of B from B = 100. That is a candidate change for easyRasch2 (a decision outside this revision), and it would mean the 400-iteration floor is needed for reproducibility rather than for calibration.

E3 and Study 1: interval convergence from fresh data

Width relative to the width at B = 10 000 of the same bootstrap, median over items and replications, as in the submission (which used a separate bootstrap at 10 000 iterations and 30 replications per n).

```
relw_f <- H1 |>
  mutate(width = hi - lo) |>
  summarise(v = median(width), .by = c(n, w, B)) |>
  mutate(rel = v / v[B == max(B)], .by = c(n, w))
CV <- do.call(rbind, lapply(seq_len(nrow(s_conv$grid)), function(i) {
  r <- s_conv$res[[i]]
  if (is.null(r)) return(NULL)
  cbind(rep = s_conv$grid$rep[i], n = s_conv$grid$n[i], r)
}))
relw_m <- CV |>
  summarise(v = median(width), .by = c(n, w, B)) |>
  mutate(rel = v / v[B == max(B)], .by = c(n, w))
b99 <- function(relw) relw |>
  summarise(m = median(rel), .by = c(w, B)) |>
  summarise(B99 = min(B[m >= 0.99]), .by = w)
spread <- function(relw) relw |>
  summarise(sp = max(rel) - min(rel), .by = c(w, B)) |>
  summarise(mx = max(sp)) |> pull(mx)
```

```
relw_f |>
  summarise(rel = median(rel), .by = c(w, B)) |>
  pivot_wider(names_from = B, values_from = rel, names_prefix = "B = ") |>
  select(-`B = 10000`) |>
  knitr::kable(digits = 3)
```

w	B = 100	B = 200	B = 400	B = 1000	B = 2000	B = 5000
0.900	0.966	0.978	0.984	0.993	0.995	0.998
0.950	0.963	0.976	0.986	0.991	0.996	1.000
0.990	0.925	0.956	0.980	0.989	0.994	0.998
0.995	0.891	0.938	0.965	0.986	0.994	0.998
0.999	0.762	0.839	0.907	0.962	0.981	0.996

```
left_join(b99(relw_f) |> rename(`B99 fresh` = B99),
          b99(relw_m) |> rename(`B99 submitted` = B99), by = "w") |>
  knitr::kable()
```

w	B99 fresh	B99 submitted
0.900	1000	1000

w	B99 fresh	B99 submitted
0.950	1000	1000
0.990	2000	2000
0.995	2000	2000
0.999	5000	5000

```
data.frame(data = c("fresh", "submitted"),
           max_spread_between_n = c(spread(relw_f), spread(relw_m))) |>
  knitr::kable(digits = 3)
```

data	max_spread_between_n
fresh	0.008
submitted	0.018

Iterations reaching 99% of the reference width, and the largest spread between sample sizes

Summary for Study 1. The convergence results replicate in fresh data. The iterations needed to reach 99% of the reference width are the same as submitted for every width (1000 for .90 and .95, 2000 for .99 and .995, 5000 for .999), the .999 interval reaches 0.76 of the reference at B = 100 as before, and the largest spread between sample sizes is smaller (0.008 against 0.018), with 500 replications per n instead of 30. The reference is now B = 10 000 of the same bootstrap rather than a separate one, which the Method must say.

Interval rule under the null against Equation 1

```
S1 |>
  filter(grepl("^Interval", rule)) |>
  mutate(w = as.numeric(sub("Interval ", "", rule))) |>
  select(w, B, FWER) |>
  pivot_wider(names_from = B, values_from = FWER, names_prefix = "B = ") |>
  mutate(pred = 100 * (1 - w^20)) |>
  arrange(w) |>
  mutate(w = sprintf("%.3f", w)) |>
  knitr::kable(digits = 1)
```

w	B = 100	B = 200	B = 400	B = 1000	B = 2000	B = 5000	B = 10000	pred
0.900	91.0	89.8	88.9	87.9	87.0	86.1	85.9	87.8
0.950	71.9	68.2	65.8	63.6	62.9	61.6	61.1	64.2
0.990	37.0	28.1	22.4	18.6	17.8	16.4	16.0	18.2
0.995	30.1	19.2	14.5	9.7	8.8	8.2	7.6	9.5
0.999	30.1	15.7	7.7	3.4	2.6	2.0	1.6	2.0

```
bind_part(s_grid, "hdci", keys) |>
  filter(k == 0) |>
  summarise(any = any(code > 0), .by = c(rep, n, B, w)) |>
  summarise(FWER = 100 * mean(any), .by = c(w, B)) |>
  pivot_wider(names_from = B, values_from = FWER, names_prefix = "B = ") |>
  mutate(pred = 100 * (1 - w^20)) |>
  arrange(w) |>
```

```
  mutate(w = sprintf("%.3f", w)) |>
  knitr::kable(digits = 1)
```

w	B = 100	B = 200	B = 400	B = 1000	B = 2000	pred
0.900	91.7	88.6	88.2	86.8	87.2	87.8
0.950	74.1	66.7	66.3	62.9	61.9	64.2
0.990	35.8	29.7	23.3	21.0	18.5	18.2
0.995	29.7	20.8	14.2	12.2	10.3	9.5
0.999	29.7	17.2	8.6	4.6	2.9	2.0

Summary. The interval rule behaves as submitted. At w = .999 its familywise error was 30.1% at B = 100, 7.7% at 400 and 2.6% at 2000, against 29.7, 8.6 and 2.9% submitted, and it reaches 1.6% at 10 000, against the Equation 1 prediction of 2.0%. At B = 10 000 every width is at or below its prediction (ratios 0.80 to 0.98), consistent with mild positive dependence and with the conservatism found for the p-value rules. For widths .90 to .99 at B = 2000 the ratio of observed to predicted was 0.98 to 0.99.

Study 2 misfit conditions from fresh data

The detection and false positive trade-off (section 4.3)

```
trade <- function(L) L |>
  filter(k == 3, n == 150, rule == "Interval 0.999") |>
  summarise(Det = 100 * mean(flag[is_mis]), FP = 100 * mean(flag[!is_mis]),
            .by = B) |> arrange(B)
art_m <- bind_part(s_grid, "hdci", keys) |>
  filter(k == 3, n == 150, abs(w - 0.999) < 1e-9) |>
  mutate(is_mis = item %in% s_grid$misfit) |>
  summarise(Det = 100 * mean(code[is_mis] > 0), FP = 100 * mean(code[!is_mis] > 0),
            .by = B) |> arrange(B)
left_join(trade(L2), art_m, by = "B", suffix = c(" fresh", " submitted")) |>
  knitr::kable(digits = 1)
```

B	Det fresh	FP fresh	Det submitted	FP submitted
100	83.9	4.2	80.8	4.0
200	78.8	2.7	75.2	2.2
400	72.4	1.4	68.7	1.3
1000	66.5	0.5	65.3	0.5
2000	67.2	0.5	64.9	0.4

```
L2 |>
  filter(k == 1, n == 150, rule == "Interval 0.999", is_mis) |>
  summarise(Det = 100 * mean(flag), .by = B) |> arrange(B) |>
  knitr::kable(digits = 1)
```

B	Det
100	100.0
200	99.6

B	Det
400	99.6
1000	99.6
2000	99.6

Reachability sequences (section 5.1)

```
summ(L2 |> filter(n == 150), "k") |>
  filter(rule %in% c("Holm", "Westfall-Young", "Benjamini-Hochberg")) |>
  select(k, rule, B, Det) |>
  pivot_wider(names_from = B, values_from = Det, names_prefix = "B = ") |>
  arrange(k, rule) |>
  knitr::kable(digits = 1)
```

k	rule	B = 100	B = 200	B = 400	B = 1000	B = 2000
1	Benjamini-Hochberg	0.0	6.8	99.6	99.6	100.0
1	Holm	0.0	0.0	99.6	99.6	100.0
1	Westfall-Young	99.6	100.0	100.0	100.0	100.0
3	Benjamini-Hochberg	17.5	75.9	82.8	82.4	84.1
3	Holm	0.0	0.0	70.8	71.9	75.9
3	Westfall-Young	77.3	77.9	77.6	76.9	77.2

The submission reported, at n = 150, Benjamini-Hochberg detection of 0, 8, 98, 99 and 100% (one misfitting item) and 14, 71, 78, 79 and 80% (three), and Westfall-Young detection of 100% and 74% at every B.

Partial nulls: familywise error over fitting items and FDR (R1.1, R1.3)

```
S2 <- summ(L2, c("k", "n"))
show_rules <- c("Interval 0.999", "Marginal", "Holm", "Westfall-Young",
                "Benjamini-Hochberg")
S2 |>
  filter(B == 2000, rule %in% show_rules) |>
  select(k, n, rule, FWER) |>
  pivot_wider(names_from = rule, values_from = FWER) |>
  rename(Interval = `Interval 0.999`, WY = `Westfall-Young`,
         BH = `Benjamini-Hochberg`) |>
  arrange(k, n) |>
  knitr::kable(digits = 1)
```

k	n	Marginal	Holm	WY	BH	Interval
1	150	59.6	2.4	2.4	5.6	4.8
1	250	72.0	3.2	4.0	10.4	6.0
1	500	86.8	9.2	11.2	18.4	10.0
1	1000	98.4	26.0	28.0	44.0	25.6
3	150	78.4	3.2	3.6	20.0	7.2
3	250	92.8	12.4	14.4	40.4	15.6
3	500	100.0	47.2	56.8	88.4	56.4

k	n	Marginal	Holm	WY	BH	Interval
3	1000	100.0	94.8	96.8	100.0	91.6

```
S2 |>
  filter(B == 2000, rule %in% show_rules) |>
  select(k, n, rule, FDR) |>
  pivot_wider(names_from = rule, values_from = FDR) |>
  rename(Interval = `Interval 0.999`, WY = `Westfall-Young`,
         BH = `Benjamini-Hochberg`) |>
  arrange(k, n) |>
  knitr::kable(digits = 1)
```

k	n	Marginal	Holm	WY	BH	Interval
1	150	33.4	1.2	1.2	2.9	2.5
1	250	41.6	1.6	2.0	5.3	3.0
1	500	55.9	4.6	5.6	9.5	5.0
1	1000	72.5	13.6	14.7	25.2	13.1
3	150	29.7	0.8	1.0	6.2	2.3
3	250	40.3	3.3	3.9	12.9	4.4
3	500	60.9	14.7	17.2	39.7	16.5
3	1000	75.4	42.9	45.7	70.0	38.0

```
L2 |>
  filter(B == 2000, is_mis, rule %in% c("Holm", "Westfall-Young")) |>
  summarise(d = mean(flag), .by = c(k, n, cell, rule)) |>
  pivot_wider(names_from = rule, values_from = d) |>
  mutate(diff = 100 * (`Westfall-Young` - Holm)) |>
  summarise(R = n(), gain = mean(diff), se = sd(diff) / sqrt(n()),
            .by = c(k, n)) |>
  arrange(k, n) |>
  knitr::kable(digits = 2)
```

k	n	R	gain	se
1	150	250	0.00	0.00
1	250	250	0.00	0.00
1	500	250	0.00	0.00
1	1000	250	0.00	0.00
3	150	250	1.33	0.59
3	250	250	0.53	0.38
3	500	250	0.00	0.00
3	1000	250	0.00	0.00

Summary for the misfit conditions. Every conclusion the submission drew from these conditions holds in fresh data, with levels a few points different.

- The trade-off of section 4.3 replicates. With three misfitting items at n = 150, interval detection at w = .999 fell from 83.9% at B = 100 to 66.5% at 1000 (submitted 80.8 to 65.3%), while false positives fell from 4.2% to 0.5% (submitted 4.0 to 0.5%). With one misfitting item detection stayed at or above 99.6%.
- The reachability sequences replicate. Benjamini-Hochberg detected 0, 6.8, 99.6, 99.6 and 100% of the single misfitting item at B = 100 to 2000 (submitted 0, 8, 98, 99, 100), and Holm shows the same floor, with 0% at B = 100 and 200. Westfall-Young was flat at 99.6 to 100% and 76.9 to 77.9% (submitted 100% and 74%).
- Holm and Westfall-Young detection differed by at most 1.3 points at B = 2000 (SE 0.6).
- Strong control fails in fresh data as it did in the cached results. At n = 1000 the Westfall-Young rate over fitting items was 28.0% with one misfitting item and 96.8% with three (cached 37.6 and 92.8%), and the Benjamini-Hochberg FDR was 25.2% and 70.0% (cached 30.8 and 67.2%). This is the strong misfit of rho = .15, so these rates are higher than at the anchored level.